VALIDATION PLAN
OPERATIONAL QUALIFICATION
CLINICAL STUDY DESIGN MODULE

JR Validated Environment — Clinical Study Design Module v1.0

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Version History

Version	Date	Author	Description
1.0	2026-07-15	Joep Rous	Initial release. Covers all 3 clinical study design scripts; 36 automated test cases.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-07-15
Reviewer	[Name]		
Approver (QA)	[Name]		

Purpose and Scope

This Operational Qualification (OQ) Validation Plan defines the test cases for the Clinical Study Design module of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability needed to demonstrate that each clinical study design script performs correctly for its intended use under 21 CFR 820.70(i) and ISO 13485, in support of clinical investigations planned under ISO 14155 / ICH E9 principles.

This plan covers 36 automated test cases executed using the pytest framework via the admin_clinical_oq runner. Test evidence is written to ~/.jrscript/<PROJECT_ID>/validation/clinical_oq_execution_<timestamp>.txt.

The three scripts compute two-arm parallel-group sample sizes: jrc_clinical_ss_means (continuous endpoint), jrc_clinical_ss_props (binary endpoint, risk difference), and jrc_clinical_ss_survival (time-to-event, log-rank). Each supports the applicable hypothesis frameworks (superiority, non-inferiority, and — except survival — equivalence/TOST) via the --framework flag, with allocation ratio, dropout inflation, and sensitivity scenario tables computed in the validated R layer. Statistical methods: Chow, Shao & Wang (2008), Sample Size Calculations in Clinical Research, 2nd ed., Chapters 3 and 4 (normal approximations); Schoenfeld (1983) events formula for the log-rank test. All three scripts use base R only.

User Requirements

The following user requirements are tested by this plan.

UR ID	Requirement
UR-CLIN-001	jrc_clinical_ss_means shall accept valid superiority inputs and exit 0 with the report header present.
UR-CLIN-002	jrc_clinical_ss_means shall compute the superiority per-arm and total sample size per the Chow ch. 3 normal-approximation formula.
UR-CLIN-003	jrc_clinical_ss_means shall report the significance level, sidedness, and the resulting z-value.
UR-CLIN-004	jrc_clinical_ss_means shall compute the non-inferiority sample size per the Chow ch. 3 margin formula at one-sided alpha.
UR-CLIN-005	jrc_clinical_ss_means shall inflate the evaluable n to an enrolled N as n/(1−dropout), rounded up per arm.
UR-CLIN-006	jrc_clinical_ss_means shall compute the equivalence sample size with the TOST beta term z(1−(1−power)/2).
UR-CLIN-007	jrc_clinical_ss_means shall support unequal allocation via --ratio (treatment:control).
UR-CLIN-008	jrc_clinical_ss_means shall print an n-vs-SD scenario table over SD × 0.8…1.2 when --sensitivity is given.
UR-CLIN-009	jrc_clinical_ss_means shall exit non-zero when superiority is requested without a meaningful difference.
UR-CLIN-010	jrc_clinical_ss_means shall exit non-zero when non-inferiority is requested without a margin.
UR-CLIN-011	jrc_clinical_ss_means shall exit non-zero when the assumed true difference is not inside the equivalence zone.
UR-CLIN-012	jrc_clinical_ss_means shall exit non-zero when --power is not strictly between 0 and 1.
UR-CLIN-013	jrc_clinical_ss_means shall exit non-zero on any unrecognized argument.
UR-CLIN-014	jrc_clinical_ss_means shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-CLIN-015	jrc_clinical_ss_props shall accept valid superiority inputs and exit 0 with the report header present.
UR-CLIN-016	jrc_clinical_ss_props shall compute the superiority sample size per the Chow ch. 4 unpooled risk-difference formula.
UR-CLIN-017	jrc_clinical_ss_props shall compute the NI sample size, defaulting the treatment rate to the control rate when --p-treat is omitted.
UR-CLIN-018	jrc_clinical_ss_props shall compute the equivalence sample size with the TOST beta term.

UR-CLIN-019	jrc_clinical_ss_props shall exit non-zero when superiority is requested without the expected treatment rate.
UR-CLIN-020	jrc_clinical_ss_props shall exit non-zero when p-treat equals p-control under superiority.
UR-CLIN-021	jrc_clinical_ss_props shall exit non-zero when non-inferiority is requested without a margin.
UR-CLIN-022	jrc_clinical_ss_props shall exit non-zero when --p-control is not strictly between 0 and 1.
UR-CLIN-023	jrc_clinical_ss_props shall print an n-vs-p-control scenario table over p-control × 0.8…1.2 when --sensitivity is given.
UR-CLIN-024	jrc_clinical_ss_props shall inflate the evaluable n to an enrolled N as n/(1−dropout), rounded up per arm.
UR-CLIN-025	jrc_clinical_ss_props shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-CLIN-026	jrc_clinical_ss_survival shall accept valid superiority inputs and exit 0 with the report header present.
UR-CLIN-027	jrc_clinical_ss_survival shall compute required events per Schoenfeld and convert to subjects via the event probability.
UR-CLIN-028	jrc_clinical_ss_survival shall compute NI events using log(margin) − log(true HR) at one-sided alpha.
UR-CLIN-029	jrc_clinical_ss_survival shall exit non-zero when --framework equivalence is requested.
UR-CLIN-030	jrc_clinical_ss_survival shall exit non-zero when superiority is requested with a hazard ratio of exactly 1.
UR-CLIN-031	jrc_clinical_ss_survival shall exit non-zero when the NI margin on the hazard ratio is ≤ 1.
UR-CLIN-032	jrc_clinical_ss_survival shall exit non-zero when the assumed true HR is not less than the margin.
UR-CLIN-033	jrc_clinical_ss_survival shall exit non-zero when the event probability is missing or out of (0, 1].
UR-CLIN-034	jrc_clinical_ss_survival shall print an n-vs-event-prob scenario table over event-prob × 0.8…1.2 when --sensitivity is given.
UR-CLIN-035	jrc_clinical_ss_survival shall inflate the evaluable n to an enrolled N as n/(1−dropout), rounded up per arm.
UR-CLIN-036	jrc_clinical_ss_survival shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.

Test Environment

All test cases are executed via the admin_clinical_oq runner using pytest with the shared OQ virtual environment at ~/.venvs/<PROJECT_ID>_oq/. Evidence is written to ~/.jrscript/<PROJECT_ID>/validation/. The scripts are parameter-only (study design happens before data exist), so the suite uses no data files.

Item	Specification
Language	R (≥ 4.3.0 recommended); base R only — no external packages
Test framework	pytest (≥ 8.0)
Test runner	admin_clinical_oq
OQ venv	~/.venvs/<PROJECT_ID>_oq/ (shared with admin_oq)
Evidence output	~/.jrscript/<PROJECT_ID>/validation/clinical_oq_execution_<timestamp>.txt
Validation plan ID	JR-VP-CLIN-001

Test Cases — jrc_clinical_ss_means

TC-CLIN-MEANS-001 Superiority happy path — exit 0

Tests UR(s)	UR-CLIN-001
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5`
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'continuous endpoint'.

TC-CLIN-MEANS-002 Superiority numeric correctness

Tests UR(s)	UR-CLIN-002
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5`
Rationale	Verifies the core superiority formula against the independently derived reference: n/arm = ceiling($2 \cdot 10^2 \cdot (z0.975+z0.80)^2/5^2$) = 63.
Pass criterion	n treatment = 63, n control = 63, n TOTAL = 126.

TC-CLIN-MEANS-003 Two-sided z-value reported

Tests UR(s)	UR-CLIN-003
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5`
Rationale	The sidedness convention is the guide's #1 pitfall; the report must state it and the z it implies.
Pass criterion	Output contains '2-sided' and z = 1.9600.

TC-CLIN-MEANS-004 Non-inferiority numeric correctness

Tests UR(s)	UR-CLIN-004
Command / action	`jrc_clinical_ss_means --framework non_inferiority --power 0.90 --alpha 0.025 --sides 1 --sd 8 --margin 4`
Rationale	Verifies the NI branch: n/arm = ceiling($2 \cdot 8^2 \cdot (z0.975+z0.90)^2/4^2$) = 85 (alpha 0.025 one-sided).
Pass criterion	n treatment = 85, n control = 85, n TOTAL = 170.

TC-CLIN-MEANS-005 Dropout inflation

Tests UR(s)	UR-CLIN-005
Command / action	`jrc_clinical_ss_means --framework non_inferiority --power 0.90 --alpha 0.025 --sides 1 --sd 8 --margin 4 --dropout 0.15`
Rationale	Enrolled N is a script responsibility (never GUI arithmetic): ceiling(85/0.85) = 100 per arm.
Pass criterion	Output contains 'ENROLL 100 + 100 = 200'.

TC-CLIN-MEANS-006 Equivalence (TOST) numeric correctness

Tests UR(s)	UR-CLIN-006
Command / action	`jrc_clinical_ss_means --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --sd 10 --margin 5`
Rationale	Verifies the TOST branch: n/arm = ceiling($2 \cdot 10^2 \cdot (z0.95+z0.90)^2/5^2$) = 69.
Pass criterion	n TOTAL = 138. Output contains 'equivalence (TOST)'.

TC-CLIN-MEANS-007 Allocation ratio 2:1

Tests UR(s)	UR-CLIN-007
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5 --ratio 2`
Rationale	Verifies the (1+1/k) allocation factor: control = ceiling($1.5 \cdot 100 \cdot 2.801585^2/25$) = 48; treatment = 95.
Pass criterion	n treatment = 95, n control = 48.

TC-CLIN-MEANS-008 Sensitivity table scales with SD²

Tests UR(s)	UR-CLIN-008
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5 --sensitivity`
Rationale	The SD is the most uncertain design input; the table must recompute n per scenario in the

	validated layer (n ∝ SD^2).
Pass criterion	5 scenario rows; totals 82 at SD=8 and 182 at SD=12; assumed row marked.

TC-CLIN-MEANS-009 Superiority requires --delta

Tests UR(s)	UR-CLIN-009
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10`
Rationale	A superiority design without delta is meaningless; the script must refuse rather than assume.
Pass criterion	Exit code ≠ 0. 'requires --delta' in combined stdout/stderr.

TC-CLIN-MEANS-010 Non-inferiority requires --margin

Tests UR(s)	UR-CLIN-010
Command / action	`jrc_clinical_ss_means --framework non_inferiority --power 0.90 --alpha 0.025 --sides 1 --sd 8`
Rationale	The NI margin is the defining design input; it must be explicit.
Pass criterion	Exit code ≠ 0. 'requires --margin' in combined stdout/stderr.

TC-CLIN-MEANS-011 Equivalence needs |delta| < margin

Tests UR(s)	UR-CLIN-011
Command / action	`jrc_clinical_ss_means --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --sd 10 --margin 5 --delta 5`
Rationale	|delta| ≥ margin makes the required n infinite; the script must refuse with a clear message.
Pass criterion	Exit code ≠ 0. '|delta| < margin' in combined stdout/stderr.

TC-CLIN-MEANS-012 --power out of range

Tests UR(s)	UR-CLIN-012
Command / action	`jrc_clinical_ss_means --framework superiority --power 1.5 --alpha 0.05 --sides 2 --sd 10 --delta 5`
Rationale	Guards against a mistyped power silently producing nonsense.
Pass criterion	Exit code ≠ 0. '--power' in combined stdout/stderr.

TC-CLIN-MEANS-013 Unknown flag rejected

Tests UR(s)	UR-CLIN-013
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5 --bogus 1`
Rationale	A typo in a flag name must never be silently ignored in a validated tool.
Pass criterion	Exit code ≠ 0. 'Unknown argument' in combined stdout/stderr.

TC-CLIN-MEANS-014 Bypass protection

Tests UR(s)	UR-CLIN-014
Command / action	`Rscript repos/clinical/R/jrc_clinical_ss_means.R ... (RENV_PATHS_ROOT unset)`
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_ss_props

TC-CLIN-PROPS-001 Superiority happy path — exit 0

Tests UR(s)	UR-CLIN-015
Command / action	`jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70 --p-treat 0.85`
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'binary endpoint'.

TC-CLIN-PROPS-002 Superiority numeric correctness

Tests UR(s)	UR-CLIN-016
Command / action	`jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70 --p-treat 0.85`
Rationale	Verifies the core formula: V = 0.85·0.15 + 0.70·0.30 = 0.3375; n/arm = ceiling(0.3375·(z0.975+z0.80)²/0.15²) = 118.
Pass criterion	n treatment = 118, n control = 118, n TOTAL = 236.

TC-CLIN-PROPS-003 Non-inferiority numeric, default p-treat

Tests UR(s)	UR-CLIN-017
Command / action	`jrc_clinical_ss_props --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --p-control 0.85 --margin 0.10`
Rationale	Verifies the NI branch and the documented default: V = 2·0.85·0.15 = 0.255; n/arm = ceiling(0.255·7.848886/0.01) = 201.
Pass criterion	n/arm = 201, n TOTAL = 402; 'assumed equal to control' in output.

TC-CLIN-PROPS-004 Equivalence (TOST) numeric correctness

Tests UR(s)	UR-CLIN-018
Command / action	`jrc_clinical_ss_props --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --p-control 0.80 --margin 0.15`
Rationale	Verifies the TOST branch: V = 2·0.80·0.20 = 0.32; n/arm = ceiling(0.32·(z0.95+z0.90)²/0.15²) = 122.
Pass criterion	n TOTAL = 244. Output contains 'equivalence (TOST)'.

TC-CLIN-PROPS-005 Superiority requires --p-treat

Tests UR(s)	UR-CLIN-019
Command / action	`jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70`
Rationale	A superiority design needs the treatment rate; the NI/equivalence default must not leak into superiority.
Pass criterion	Exit code ≠ 0. 'requires --p-treat' in combined stdout/stderr.

TC-CLIN-PROPS-006 Equal rates rejected for superiority

Tests UR(s)	UR-CLIN-020
Command / action	`jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70 --p-treat 0.70`
Rationale	Zero effect size gives an infinite n; the script must refuse.
Pass criterion	Exit code ≠ 0. 'different from p-control' in combined stdout/stderr.

TC-CLIN-PROPS-007 Non-inferiority requires --margin

Tests UR(s)	UR-CLIN-021
Command / action	`jrc_clinical_ss_props --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --p-control 0.85`
Rationale	The NI margin is the defining design input; it must be explicit.
Pass criterion	Exit code ≠ 0. 'requires --margin' in combined stdout/stderr.

TC-CLIN-PROPS-008 --p-control out of range

Tests UR(s)	UR-CLIN-022
Command / action	`jrc_clinical_ss_props --framework non_inferiority --power 0.80 --`

	`alpha 0.025 --sides 1 --p-control 1.2 --margin 0.10`
Rationale	Guards against an impossible rate silently producing nonsense.
Pass criterion	Exit code ≠ 0. '--p-control' in combined stdout/stderr.

TC-CLIN-PROPS-009 Sensitivity table over the control rate

Tests UR(s)	UR-CLIN-023
Command / action	`jrc_clinical_ss_props --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --p-control 0.80 --margin 0.15 --sensitivity`
Rationale	The control rate is the most uncertain design input; scenarios must be recomputed in the validated layer.
Pass criterion	5 scenario rows (0.64…0.96); assumed row marked.

TC-CLIN-PROPS-010 Dropout inflation

Tests UR(s)	UR-CLIN-024
Command / action	`jrc_clinical_ss_props --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --p-control 0.85 --margin 0.10 --dropout 0.10`
Rationale	Enrolled N is a script responsibility: ceiling(201/0.90) = 224 per arm.
Pass criterion	Output contains 'ENROLL 224 + 224 = 448'.

TC-CLIN-PROPS-011 Bypass protection

Tests UR(s)	UR-CLIN-025
Command / action	`Rscript repos/clinical/R/jrc_clinical_ss_props.R ... (RENV_PATHS_ROOT unset)`
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_ss_survival

TC-CLIN-SURV-001 Superiority happy path — exit 0

Tests UR(s)	UR-CLIN-026
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6`
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'time-to-event endpoint'.

TC-CLIN-SURV-002 Superiority numeric correctness

Tests UR(s)	UR-CLIN-027
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6`
Rationale	Verifies the events formula: E = ceiling(4·(z0.975+z0.80)²/log(0.70)²) = 247; n/arm = ceiling((246.86/0.6)/2) = 206.
Pass criterion	Events required = 247, n treatment = 206, n control = 206, n TOTAL = 412.

TC-CLIN-SURV-003 Non-inferiority numeric correctness

Tests UR(s)	UR-CLIN-028
Command / action	`jrc_clinical_ss_survival --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --hr 1.0 --margin 1.3 --event-prob 0.5`
Rationale	Verifies the NI branch: E = ceiling(4·7.848886/log(1.3)²) = 457; n/arm = ceiling((456.10/0.5)/2) = 457.
Pass criterion	Events required = 457, n TOTAL = 914.

TC-CLIN-SURV-004 Equivalence not offered

Tests UR(s)	UR-CLIN-029
Command / action	`jrc_clinical_ss_survival --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --hr 1.0 --event-prob 0.5`
Rationale	Equivalence is deliberately out of scope for time-to-event endpoints in this module.
Pass criterion	Exit code ≠ 0. 'not offered' in combined stdout/stderr.

TC-CLIN-SURV-005 Superiority with HR = 1 rejected

Tests UR(s)	UR-CLIN-030
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 1 --event-prob 0.6`
Rationale	HR = 1 means no effect; required events would be infinite.
Pass criterion	Exit code ≠ 0. 'different from 1' in combined stdout/stderr.

TC-CLIN-SURV-006 NI margin must exceed 1

Tests UR(s)	UR-CLIN-031
Command / action	`jrc_clinical_ss_survival --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --hr 1.0 --margin 0.9 --event-prob 0.5`
Rationale	An NI margin ≤ 1 is not a non-inferiority margin on a hazard ratio.
Pass criterion	Exit code ≠ 0. '--margin' in combined stdout/stderr.

TC-CLIN-SURV-007 NI needs true HR inside the margin

Tests UR(s)	UR-CLIN-032
Command / action	`jrc_clinical_ss_survival --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --hr 1.4 --margin 1.3 --event-prob 0.5`
Rationale	A true HR at or beyond the margin cannot demonstrate non-inferiority at any n.
Pass criterion	Exit code ≠ 0. '--hr < --margin' in combined stdout/stderr.

TC-CLIN-SURV-008 --event-prob required

Tests UR(s)	UR-CLIN-033
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70`

Rationale	Events cannot be converted to subjects without the event probability.
Pass criterion	Exit code ≠ 0. '--event-prob' in combined stdout/stderr.

TC-CLIN-SURV-009 Sensitivity table over the event probability

Tests UR(s)	UR-CLIN-034
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6 --sensitivity`
Rationale	The event probability is the most uncertain design input; scenarios must be recomputed in the validated layer.
Pass criterion	5 scenario rows (0.48…0.72); assumed row marked.

TC-CLIN-SURV-010 Dropout inflation

Tests UR(s)	UR-CLIN-035
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6 --dropout 0.20`
Rationale	Enrolled N is a script responsibility: ceiling(206/0.80) = 258 per arm.
Pass criterion	Output contains 'ENROLL 258 + 258 = 516'.

TC-CLIN-SURV-011 Bypass protection

Tests UR(s)	UR-CLIN-036
Command / action	`Rscript repos/clinical/R/jrc_clinical_ss_survival.R ... (RENV_PATHS_ROOT unset)`
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Requirements Traceability Matrix

Maps each test case to the user requirement(s) it covers.

Test Case ID	User Requirement(s)
TC-CLIN-MEANS-001	UR-CLIN-001
TC-CLIN-MEANS-002	UR-CLIN-002
TC-CLIN-MEANS-003	UR-CLIN-003
TC-CLIN-MEANS-004	UR-CLIN-004
TC-CLIN-MEANS-005	UR-CLIN-005
TC-CLIN-MEANS-006	UR-CLIN-006
TC-CLIN-MEANS-007	UR-CLIN-007
TC-CLIN-MEANS-008	UR-CLIN-008
TC-CLIN-MEANS-009	UR-CLIN-009
TC-CLIN-MEANS-010	UR-CLIN-010
TC-CLIN-MEANS-011	UR-CLIN-011
TC-CLIN-MEANS-012	UR-CLIN-012
TC-CLIN-MEANS-013	UR-CLIN-013
TC-CLIN-MEANS-014	UR-CLIN-014
TC-CLIN-PROPS-001	UR-CLIN-015
TC-CLIN-PROPS-002	UR-CLIN-016
TC-CLIN-PROPS-003	UR-CLIN-017
TC-CLIN-PROPS-004	UR-CLIN-018
TC-CLIN-PROPS-005	UR-CLIN-019
TC-CLIN-PROPS-006	UR-CLIN-020
TC-CLIN-PROPS-007	UR-CLIN-021
TC-CLIN-PROPS-008	UR-CLIN-022
TC-CLIN-PROPS-009	UR-CLIN-023
TC-CLIN-PROPS-010	UR-CLIN-024
TC-CLIN-PROPS-011	UR-CLIN-025
TC-CLIN-SURV-001	UR-CLIN-026
TC-CLIN-SURV-002	UR-CLIN-027
TC-CLIN-SURV-003	UR-CLIN-028
TC-CLIN-SURV-004	UR-CLIN-029
TC-CLIN-SURV-005	UR-CLIN-030
TC-CLIN-SURV-006	UR-CLIN-031
TC-CLIN-SURV-007	UR-CLIN-032
TC-CLIN-SURV-008	UR-CLIN-033
TC-CLIN-SURV-009	UR-CLIN-034
TC-CLIN-SURV-010	UR-CLIN-035
TC-CLIN-SURV-011	UR-CLIN-036